CIUK 2026, UCL challenge

# Which patients' E. coli infections resist co-amoxiclav?

Tree models, run on UCL's Condenser cloud

**UCLuster Too**

Ada Ge · Bobby Ta · Eric Huang · James Loke

**Q1.** What is cloud compute?

# Cloud computing: renting computers over the internet when you need them

| | |
|---|---|
| **Our code** | **we run this** |
| **Our virtual machine: a computer made in software** | Condenser gave us this |
| **Hardware shared with other users** | **UCL runs this** |

Condenser is UCL's private cloud for research computing. This split is called infrastructure as a service.

**Q2.** What are key data security considerations for cloud computing?

# Security on Condenser is split between UCL and us

### UCL

- Hardware in UCL's own facilities
- Each team's machines kept apart from other teams'

### Us

- Log in only through UCL's JupyterHub, with an SSH key, over encrypted connections
- No passwords or keys in our public repository

> Condenser's own page: "not certified for the secure storage and processing of sensitive data"

**Q3.** The data file was world-readable. Why is this not desirable? Did we change it, how and why?

# World-readable means anyone on the machine can read patient records, so we locked the file to our account

```
# before: any user on the machine can read the file
-rw-r--r--. 1 cc cc 94198 Oct 26 16:44 dig_health_hub_amr.csv

# moved it into data/, then
$ chmod 700 data
$ chmod 600 data/dig_health_hub_amr.csv

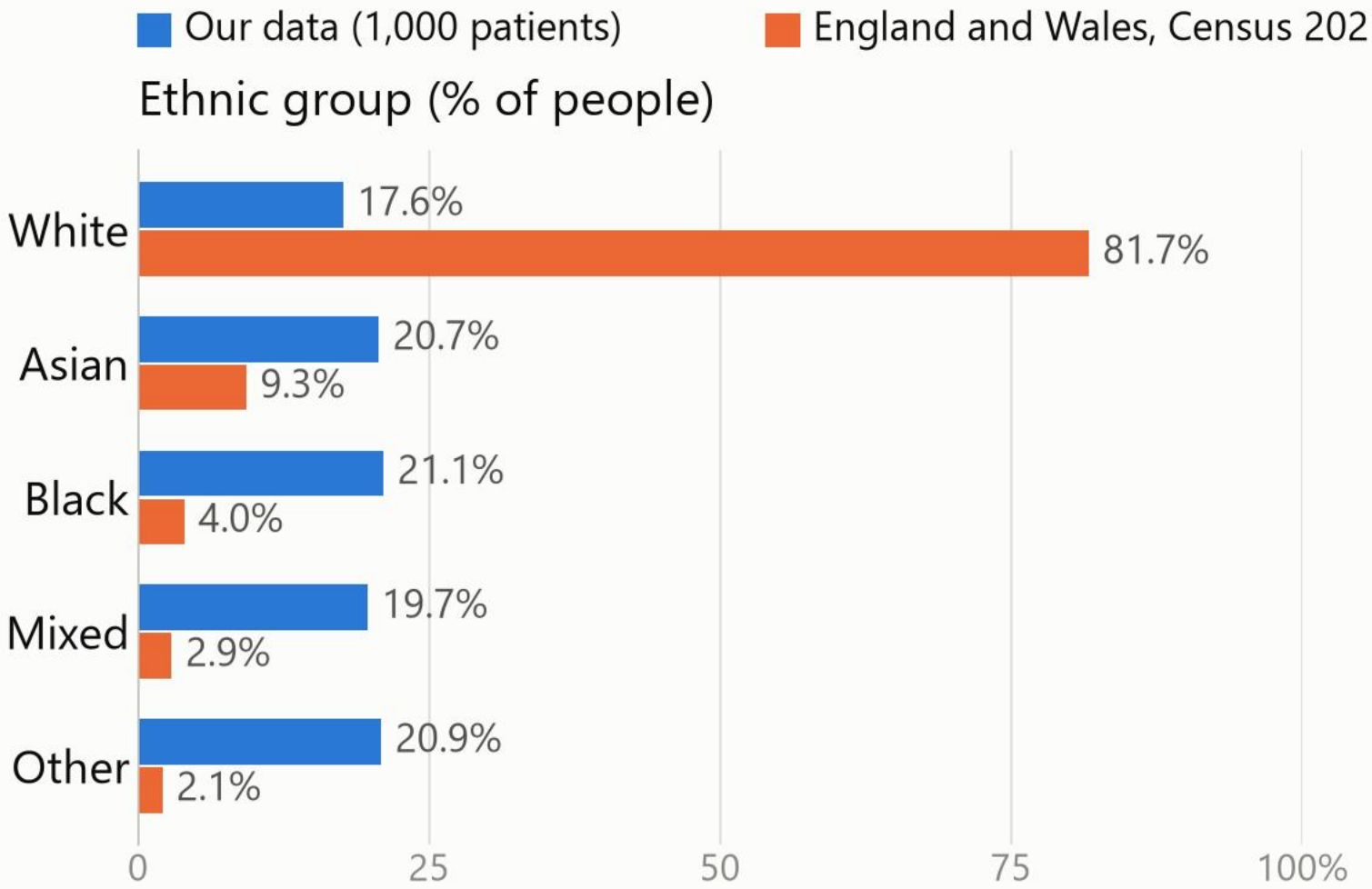
# after: only our account (cc) can open the folder or read the file
drwx------. 2 cc cc    36 Oct 27 09:51 data
-rw-------. 1 cc cc 94198 Oct 26 16:44 dig_health_hub_amr.csv
```

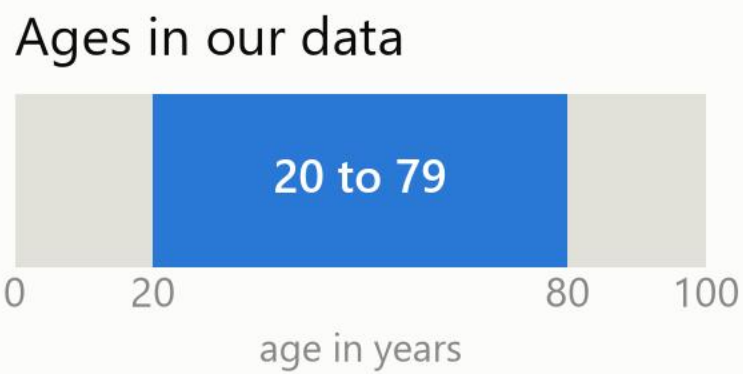
Our scripts also set `umask 027`, so the files they write are closed to other users.

**Q4.** Why would you want a representative data set for machine learning?

# A model only learns the patients it's shown

Ours has far fewer White patients than England and Wales, and no one under 20 or over 79




Nobody under 20 or over 79, so the model knows nothing about them

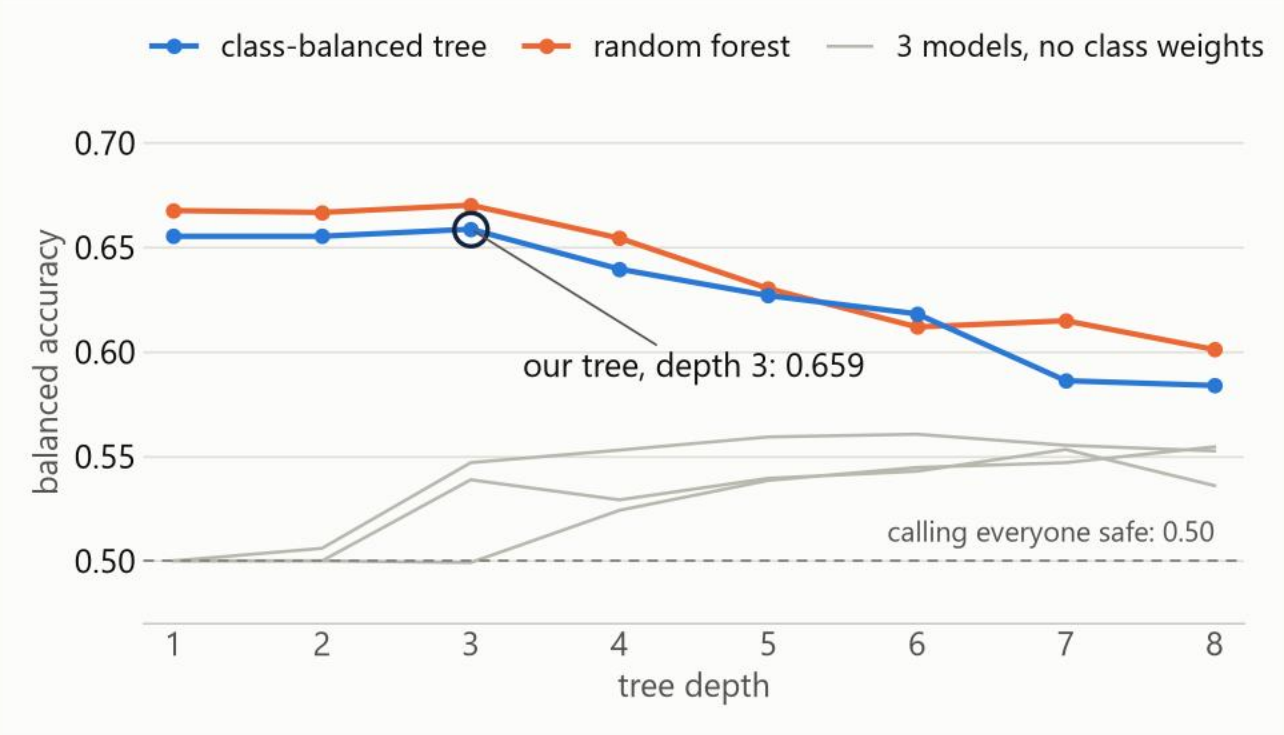
**Q5.** Why would we want to use synthetic data instead of actual patient data?

# Synthetic data lets us build models without putting real patients' records at risk

| | |
|---|---|
| **Real records** | Protected by UK GDPR. Patients may not have agreed to this use. |
| **Synthetic data** | Far less privacy risk, even on a shared cloud machine. |

**Q6.** What is our final tree model, and why is it the most optimal?

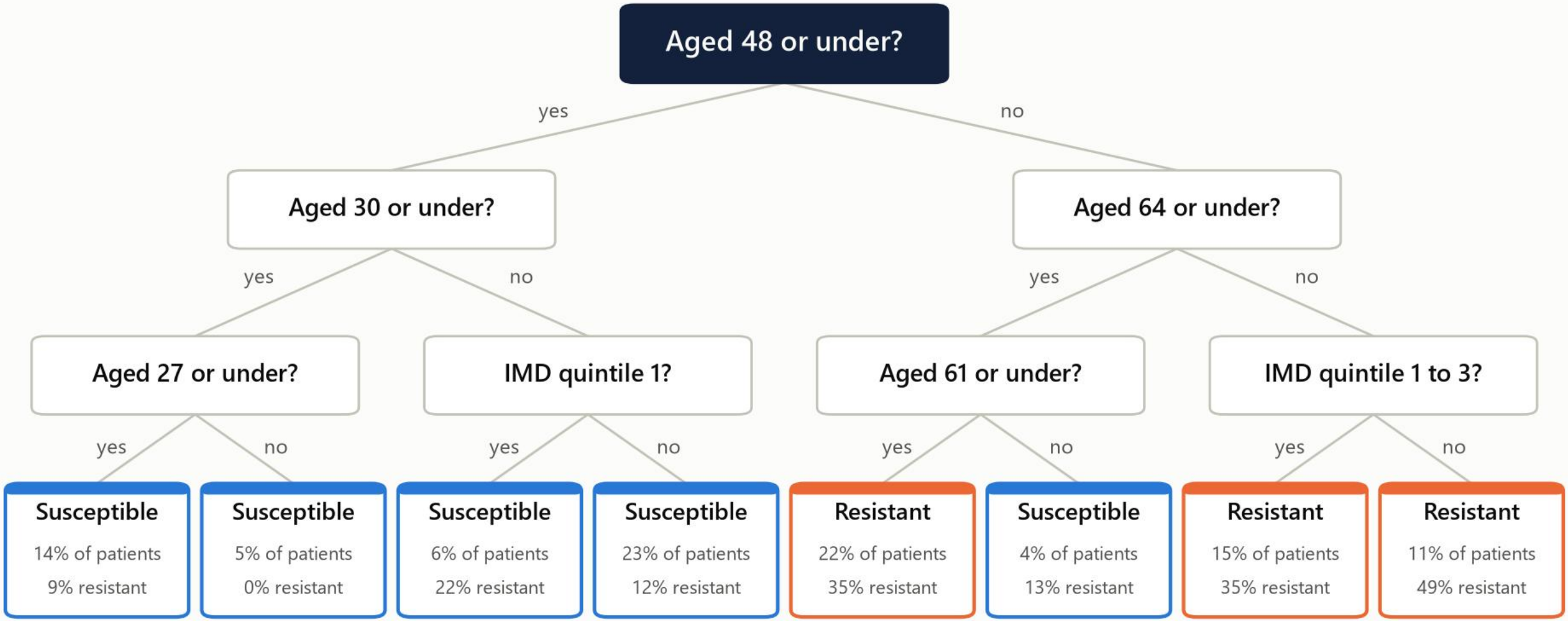
# Depth 3 gave the best balanced accuracy of any single tree




| Balanced accuracy | |
|---|---|
| Calling everyone safe (76% accuracy) | 0.50 |
| **Our tree, depth 3** | **0.659** |
| Random forest, 200 trees | 0.670 |

# Our final model: a class-balanced decision tree, depth 3

End boxes: share of patients, and how many of them were resistant. IMD quintile 1 = most deprived.

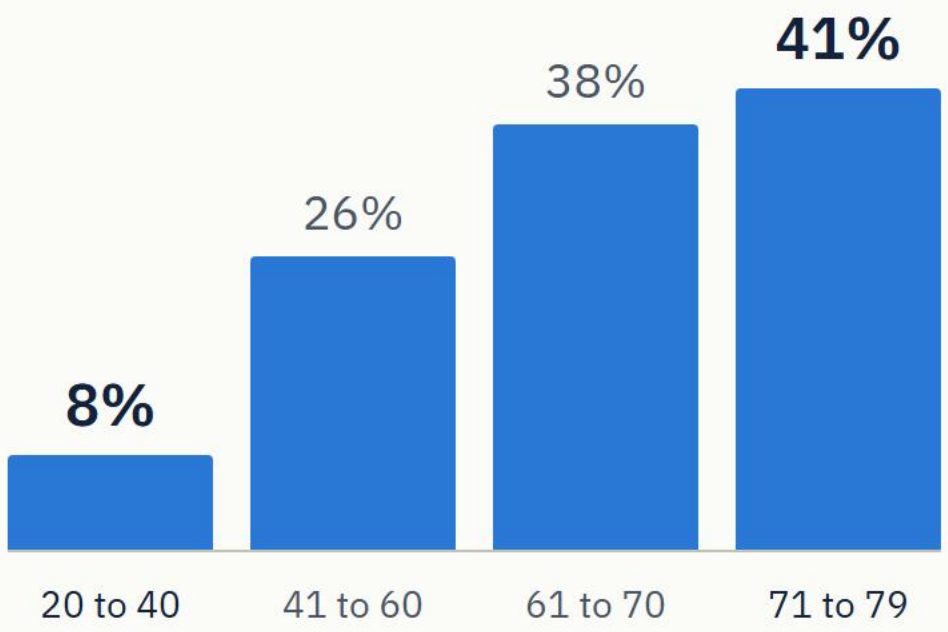


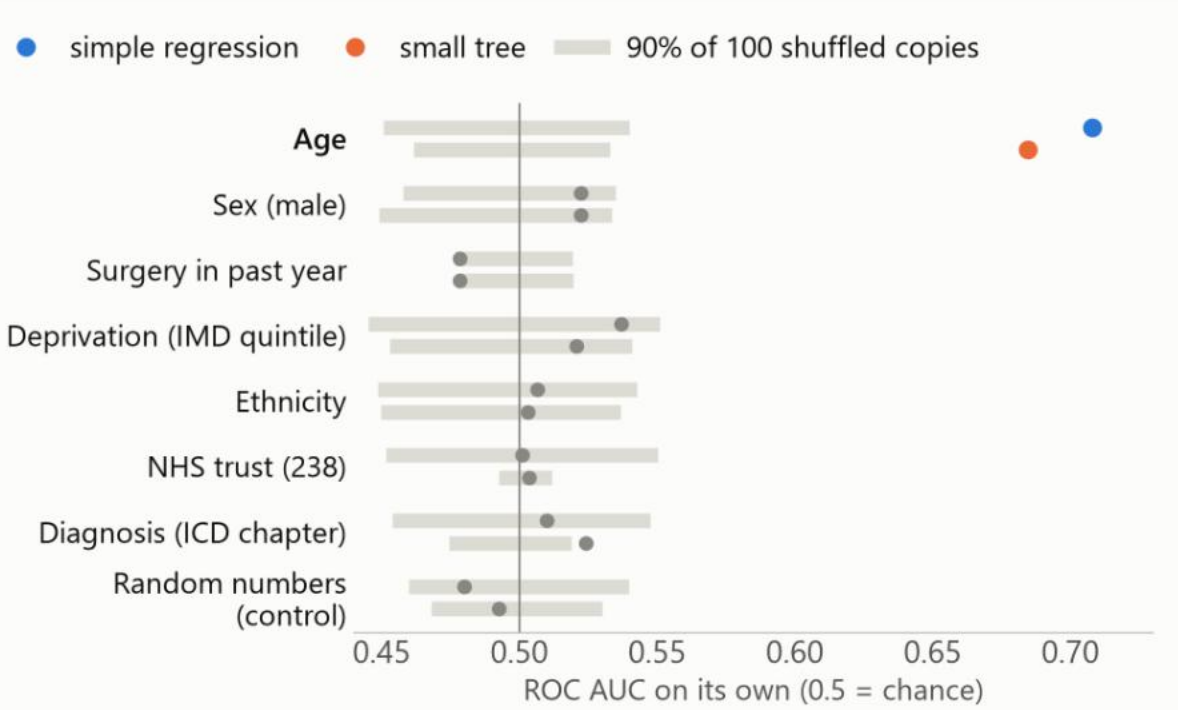

**Read as a rule: over 48, flagged resistant, except ages 62 to 64. Aged 48 or under, susceptible.**

**Q6.** What is our final tree model, and why is it the most optimal?

# Older patients are most at risk, and no other factor beat noise

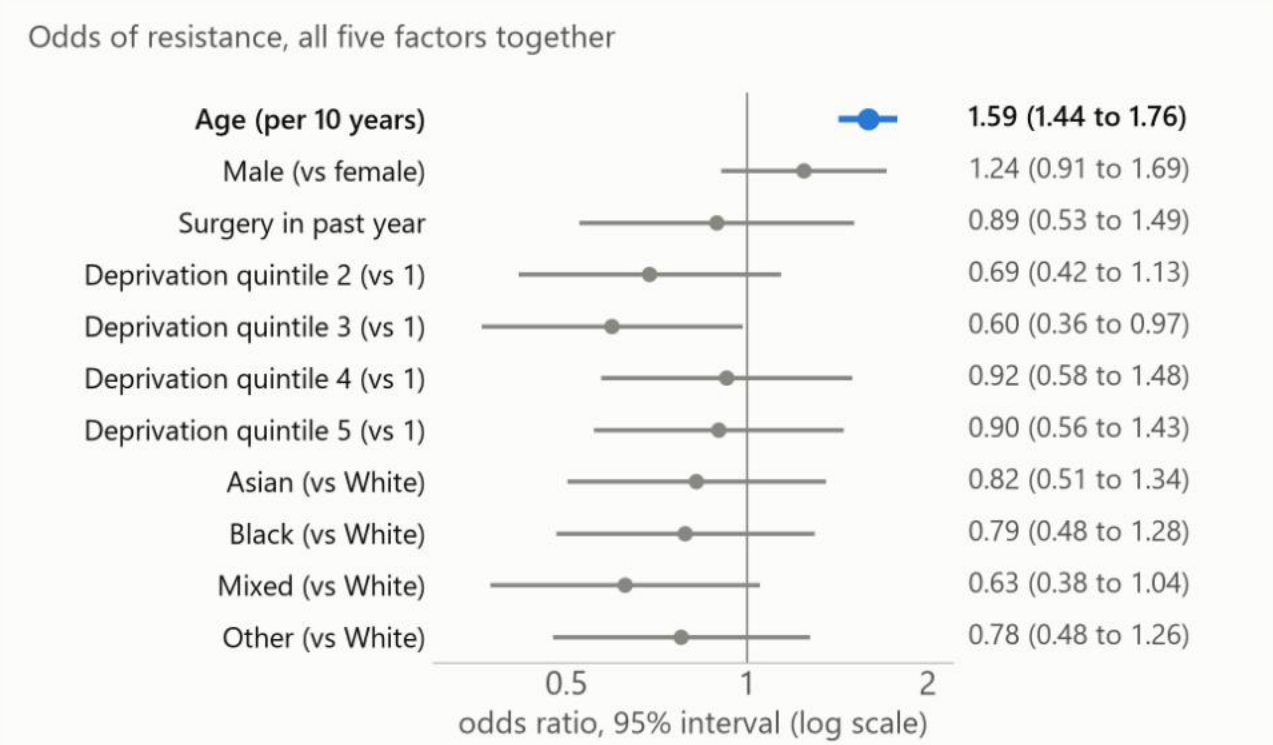
**Resistant, by age**



**Q7.** What other modelling techniques could we use, and why?

# Logistic regression also works: it matched our tree




Balanced accuracy

| | |
|---|---|
| Logistic regression | **0.660** |
| Our tree | **0.659** |

Fits risk that rises smoothly with age, and gives odds ratios. A neural network isn't needed for 1,000 patients and one clear risk factor.

**Q8.** How did we work as a team?

# Who did what

| | |
|---|---|
| **Bobby** | Cloud machine, environment and ReadMe |
| **James** | Model script |
| **Ada** | Explored the data and chose how to evaluate the models |
| **Eric** | Led the video |

We paired up to cover each other and checked in for 15 minutes each evening. The plan, role guides and work-session log are in our GitHub repository.

**Q8.** How did we work as a team?

# Problems we ran into

| Problem | What we did |
| --- | --- |
| Login details reached only one of us, with no username | Shared that login as instructed, and found the username in the organisers' access guide |
| No git on the cloud machine | Edited on laptops, pushed to GitHub, fetched files with curl |
| Our first model flagged nobody as resistant | Switched to balanced accuracy and class weights |

# Age is the only clear risk factor

Final model: a class-balanced decision tree, depth 3, balanced accuracy 0.66

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`github.com/adellama75/cluster-challenge-ucl`